

M5 Drawdown Circuit Breaker + M7 Regime Filter

Final Presentation

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Hypothesis

Objective: Combat capital erosion and improve risk adjusted returns particularly during periods of stress.

Combined Hypothesis: Integrating broad market trend filtering (M7) with portfolio capital failsafes (M5) creates a proactive/reactive shield, aiming to reduce Max Drawdown and enhancing risk-adjusted returns (Calmar/Sortino).

The Why?: Backtests showed the algorithm fighting the macro trend, so we added a SPY Regime Filter to eliminate counter-trend trades. We implemented a Circuit Breaker as a hard failsafe to automatically cut exposure and preserve capital during severe drawdowns.

Implementation

Regime Filter (SPY Trend)

signals/alpha.py: Track SPY's 200-day SMA. **Inside_compute_signals()**, suppress long signals if SPY is below the SMA, and suppress short signals if SPY is above it.

main.py: Add SPY to the algorithm's universe so the alpha model has access to the broad market data.

Circuit Breaker (NAV Drawdown)

main.py: Inside **OnData()**, track the portfolio's highest peak NAV. If the current drawdown hits -10%, calculate a reduction multiplier becomes 0.0 and we go flat and pass it to the risk module.

risk/portfolio.py: **InsideCreateTargets()**, multiply all target position weights by that drawdown multiplier before the orders are sent to the market.

Results

Metric	Strategy	2012-2014 (N1)	2016-2018 (N2)	2007-2009 (C1)	2020-2022 (C2)
Total Return	Baseline	13.41%	13.80%	7.94%	(5.51%)
Total Return	Modified	11.96%	11.08%	12.61%	6.57%
Sharpe Ratio	Baseline	0.856	0.720	(0.078)	(0.201)
Sharpe Ratio	Modified	0.855	0.693	0.059	0.278
Sortino Ratio	Baseline	1.034	0.765	(0.081)	(0.208)
Sortino Ratio	Modified	1.016	0.725	0.061	0.261
Calmar Ratio	Baseline	1.270	0.720	0.360	(0.180)
Calmar Ratio	Modified	1.304	0.733	0.736	0.267
Max Drawdown	Baseline	5.10%	8.80%	10.80%	15.90%
Max Drawdown	Modified	4.40%	7.50%	8.60%	12.10%

Deep Dive

Interesting Period: COVID Crisis

Why this window matters: Universe is DOW 30 — every name is in SPY, so the 200-day filter is a near-direct read on the book. SPY -34% in 33 days, then 22-month recovery. Baseline -5.51% → Modified +6.57%.

Why baseline failed: Pre-crash momentum was concentrated in mega-cap tech (AAPL, MSFT, V, HD) — held long into the collapse. Counter-trend rallies of +6–9% in Feb–Mar 2020 whipsawed re-entries within 24–48 hours.

M7 fixed direction: SPY broke 200-day in late Feb 2020 → longs suppressed during crash. Recrossed in late May → shorts suppressed for ~20 of 24 months, stopping the strategy from fighting squeezes in BA, CVX, JPM.

M5 fixed magnitude: With only 30 names, concentration risk is real. At -10% drawdown, weights scale down so capital actually exits. Drove max drawdown 15.9% → 12.1%

Crisis vs Normal

Δ vs. Baseline	N1 (2012–14)	N2 (2016–18)	C1 (2007–09)	C2 (2020–22)
Total Return	−1.5 pp	−2.7 pp	+4.7 pp	+12.1 pp
Sharpe Ratio	≈ 0	−0.03	+0.14	+0.48
Max Drawdown	−0.7 pp	−1.3 pp	−2.2 pp	−3.8 pp

pp = percentage points

Interaction: In normal regimes, both overlays sit inactive and the strategy trades like baseline. In crises, SPY breaks its 200-day and drawdown trips −10% early — both overlays activate together.

Amplify or dampen: Dampens stress in both directions — M7 cuts directional exposure to a falling tape, M5 cuts gross exposure once losses compound. Does not dampen recovery participation.

Robustness: Every period now has positive Sharpe and Calmar. Return spread compresses from ~19 pp to ~6 pp — ~2 pp of upside in calm markets traded for ~12 pp of crisis protection.

Limitations

Lagging regime signal: The 200-day SMA is reactive by construction. The strategy missed roughly the first two months of the 2020 recovery while waiting for SPY to recross. In a sharper V-shaped recovery, the lag is purely a cost.

Limited sample, narrow regimes: Four 2-year windows $\approx$ 8 years of disjoint data. Differences between Sharpes are likely not statistically distinguishable. Strategy is untested in sustained sideways markets where SPY repeatedly crosses its 200-day.

Survivorship-biased universe: DOW 30 is composed entirely of large-cap blue chips that have already survived. No exposure to delistings, bankruptcies, or true single-name tail risk. Average pairwise correlation is high, so 30 names provides far less diversification than the count suggests.

Cost: Because of more trades than the baseline strategy this modification is costlier.

Thank You

Q&A